

Internal Dependencies

References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [Neo4j Python Driver](#)

Artifacts

List the artifacts this notebook is based on. Different sorting variations help finding artifacts by their features and support larger code bases where the list of all artifacts gets too long.

Only the top 30 entries are shown. The whole table can be found in the following CSV report:

List_all_Java_artifacts

Table 1a - Top 30 artifacts with the highest package count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-messaging-5.0.3.jar	59	579	7	2
1	axon-common-5.0.3.jar	15	156	10	0
2	axon-eventsourcing-5.0.3.jar	7	100	3	4
3	axon-modelling-5.0.3.jar	7	92	2	3
4	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
5	axon-test-5.0.3.jar	6	78	1	3
6	axon-conversion-5.0.3.jar	5	35	4	1
7	axon-server-connector-5.0.3.jar	5	72	1	4
8	axon-update-5.0.3.jar	5	23	1	1
9	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
10	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2

Table 1b - Top 30 artifacts with the highest type count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-messaging-5.0.3.jar	59	579	7	2
1	axon-common-5.0.3.jar	15	156	10	0
2	axon-eventsourcing-5.0.3.jar	7	100	3	4
3	axon-modelling-5.0.3.jar	7	92	2	3
4	axon-test-5.0.3.jar	6	78	1	3
5	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
6	axon-server-connector-5.0.3.jar	5	72	1	4
7	axon-conversion-5.0.3.jar	5	35	4	1
8	axon-update-5.0.3.jar	5	23	1	1
9	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
10	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2

Table 1c - Top 30 artifacts with the highest number of incoming dependencies

The following table lists the top 30 artifacts that are used the most by other artifacts (highest count of incoming dependencies, highest in-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-common-5.0.3.jar	15	156	10	0
1	axon-messaging-5.0.3.jar	59	579	7	2
2	axon-conversion-5.0.3.jar	5	35	4	1
3	axon-eventsourcing-5.0.3.jar	7	100	3	4
4	axon-modelling-5.0.3.jar	7	92	2	3
5	axon-server-connector-5.0.3.jar	5	72	1	4
6	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
7	axon-test-5.0.3.jar	6	78	1	3
8	axon-update-5.0.3.jar	5	23	1	1
9	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
10	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2

Table 1d - Top 30 artifacts with the highest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the highest number of other artifacts (highest count of outgoing dependencies, highest out-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
1	axon-eventsourcing-5.0.3.jar	7	100	3	4
2	axon-server-connector-5.0.3.jar	5	72	1	4
3	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
4	axon-modelling-5.0.3.jar	7	92	2	3
5	axon-test-5.0.3.jar	6	78	1	3
6	axon-messaging-5.0.3.jar	59	579	7	2
7	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2
8	axon-conversion-5.0.3.jar	5	35	4	1
9	axon-update-5.0.3.jar	5	23	1	1
10	axon-common-5.0.3.jar	15	156	10	0

Table 1e - Top 30 artifacts with the lowest package count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2
1	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
2	axon-conversion-5.0.3.jar	5	35	4	1
3	axon-server-connector-5.0.3.jar	5	72	1	4
4	axon-update-5.0.3.jar	5	23	1	1
5	axon-test-5.0.3.jar	6	78	1	3
6	axon-eventsourcing-5.0.3.jar	7	100	3	4
7	axon-modelling-5.0.3.jar	7	92	2	3
8	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
9	axon-common-5.0.3.jar	15	156	10	0
10	axon-messaging-5.0.3.jar	59	579	7	2

Table 1f - Top 30 artifacts with the lowest type count

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2
1	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
2	axon-update-5.0.3.jar	5	23	1	1
3	axon-conversion-5.0.3.jar	5	35	4	1
4	axon-server-connector-5.0.3.jar	5	72	1	4
5	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
6	axon-test-5.0.3.jar	6	78	1	3
7	axon-modelling-5.0.3.jar	7	92	2	3
8	axon-eventsourcing-5.0.3.jar	7	100	3	4
9	axon-common-5.0.3.jar	15	156	10	0
10	axon-messaging-5.0.3.jar	59	579	7	2

Table 1g - Top 30 artifacts with the lowest number of incoming dependencies

The following table lists the top 30 artifacts that are used the least by other artifacts (lowest count of incoming dependencies, lowest in-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
1	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2
2	axon-server-connector-5.0.3.jar	5	72	1	4
3	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7
4	axon-test-5.0.3.jar	6	78	1	3
5	axon-update-5.0.3.jar	5	23	1	1
6	axon-modelling-5.0.3.jar	7	92	2	3
7	axon-eventsourcing-5.0.3.jar	7	100	3	4
8	axon-conversion-5.0.3.jar	5	35	4	1
9	axon-messaging-5.0.3.jar	59	579	7	2
10	axon-common-5.0.3.jar	15	156	10	0

Table 1h - Top 30 artifacts with the lowest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the lowest number of other artifacts (lowest count of outgoing dependencies, lowest out-degree).

	artifactName	packages	types	incomingDependencies	outgoingDependencies
0	axon-common-5.0.3.jar	15	156	10	0
1	axon-conversion-5.0.3.jar	5	35	4	1
2	axon-update-5.0.3.jar	5	23	1	1
3	axon-messaging-5.0.3.jar	59	579	7	2
4	axon-tracing-opentelemetry-5.0.3.jar	1	5	0	2
5	axon-metrics-micrometer-5.0.3.jar	3	16	0	3
6	axon-modelling-5.0.3.jar	7	92	2	3
7	axon-test-5.0.3.jar	6	78	1	3
8	axon-eventsourcing-5.0.3.jar	7	100	3	4
9	axon-server-connector-5.0.3.jar	5	72	1	4
10	axon-spring-boot-autoconfigure-5.0.3.jar	7	75	1	7

Cyclic Dependencies

Cyclic dependencies occur when one package uses a class of another package and vice versa. These dependencies can lead to problems when one of these packages needs to be changed.

Table 2a - Cyclic Dependencies Overview

Show the top 40 cyclic dependencies sorted by the most promising to resolve first. This is done by calculating the number of forward dependencies (first cycle participant to second cycle participant) in relation to backward dependencies (second cycle participant back to first cycle participant). The higher this rate (approaching 1), the easier it should be to resolve the cycle by focussing on the few backward dependencies.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

[Cyclic_Dependencies](#)

Columns:

- *artifactName* identifies the artifact of the first participant of the cycle
- *packageName* identifies the package of the first participant of the cycle
- *dependentArtifactName* identifies the artifact of the second participant of the cycle
- *dependentPackageName* identifies the package of the second participant of the cycle
- *forwardToBackwardBalance* is between 0 and 1. High for many forward and few backward dependencies.
- *numberForward* contains the number of dependencies from the first participant of the cycle to the second one
- *numberBackward* contains the number of dependencies from the second participant of the cycle back to the first one
- *someForwardDependencies* lists some forward dependencies in the text format "type1 -> type2"
- *backwardDependencies* lists the backward dependencies in the format "type1 <- type2" that are recommended to get resolved

	artifactName	packageName	dependentArtifactName	dependentPackageName	forwardToBackwardBalance	numberForward	numberBack
0	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	0.959184	48	
1	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	axon-messaging-5.0.3	org.axonframework.messaging.core	0.942857	34	
2	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	0.931034	28	
3	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	0.875000	15	
4	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	0.857143	13	
5	axon-modelling-5.0.3	org.axonframework.modelling.annotation	axon-modelling-5.0.3	org.axonframework.modelling	0.846154	12	
6	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.transaction	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	0.666667	5	
7	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	0.666667	15	
8	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	0.666667	15	
9	axon-common-5.0.3	org.axonframework.common.configuration	axon-common-5.0.3	org.axonframework.common.infra	0.529412	13	
10	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	0.500000	15	
11	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	0.428571	5	
12	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.command	0.333333	4	
13	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	0.333333	2	
14	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	axon-messaging-5.0.3	org.axonframework.messaging.core.sequencing	0.333333	4	
15	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	0.200000	6	
16	axon-modelling-5.0.3	org.axonframework.modelling.entity.annotation	axon-modelling-5.0.3	org.axonframework.modelling.annotation	0.200000	3	
17	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	0.142857	8	
18	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	axon-messaging-5.0.3	org.axonframework.messaging.core	0.111111	10	

	artifactName	packageName	dependentArtifactName	dependentPackageName	forwardToBackwardBalance	numberForward	numberBack
19	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.util	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	0.000000	2	
20	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.annotation.reflection	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.annotation	0.000000	1	

Table 2b - Cyclic Dependencies Break Down

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic_Dependencies_Breakdown

Columns in addition to Table 2a:

- dependency shows the cycle dependency in the text format "type1 -> type2" (forward) or "type2<-type1" (backward)

	artifactName	packageName	dependentArtifactName	dependentPackageName	dependency	forwardToBackwardBalance	n
0	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	ChainedMessageHandlerInterceptorMember->Message	0.959184	
1	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MethodInvokingMessageHandlingMember->MessageStream\$Single	0.959184	
2	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageHandlingMember->MessageStream	0.959184	
3	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	ChainedMessageHandlerInterceptorMember->MessageStream	0.959184	
4	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	AnnotatedMessageHandlingMemberDefinition->Message	0.959184	
5	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	AnnotatedMessageHandlingMemberDefinition->MessageStream	0.959184	
6	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	AnnotationMessageTypeResolver->MessageTypeResolver	0.959184	
7	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	PayloadParameterResolver->Message	0.959184	
8	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	DefaultParameterResolverFactory->Metadata	0.959184	
9	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	DefaultParameterResolverFactory\$MessageParameterResolver->Message	0.959184	
10	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	InterceptorChainParameterResolverFactory->Message	0.959184	
11	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	AggregateTypeParameterResolverFactory <i>AggregateTypeParameterResolver</i> ->ContextResourceKey	0.959184	
12	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	InterceptorChainParameterResolverFactory->MessageStream	0.959184	
13	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	InterceptorChainParameterResolverFactory->Context\$ResourceKey	0.959184	
14	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	WrappedMessageHandlingMember->MessageStream	0.959184	
15	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	AnnotationMessageTypeResolver->MessageType	0.959184	
16	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	AnnotationMessageTypeResolver->ClassBasedMessageTypeResolver	0.959184	
17	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageStreamResolverUtils->GenericMessage	0.959184	
18	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	WrappedMessageHandlingMember->Message	0.959184	
19	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageHandlingMember->Message	0.959184	
20	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	HandlerAttributes<SimpleHandlerAttributes	0.959184	
21	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	DefaultParameterResolverFactory->Message	0.959184	
22	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageHandler->Message	0.959184	
23	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageIdentifierParameterResolverFactory\$MessageIdentifierParameterResolver->Message	0.959184	
24	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageStreamResolverUtils->MessageStream\$Empty	0.959184	
25	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	AggregateTypeParameterResolverFactory\$AggregateTypeParameterResolver->LegacyResources	0.959184	
26	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	AnnotatedHandlerInspector->Message	0.959184	
27	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MultiHandlerDefinition->MessageStream	0.959184	
28	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	AnnotatedHandlerInspector->MessageStream	0.959184	
29	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	InterceptorChainParameterResolverFactory->MessageHandlerInterceptorChain	0.959184	
30	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageStreamResolverUtils->MessageStream\$Single	0.959184	
31	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	SourceIdParameterResolverFactory\$SourceIdParameterResolver->LegacyResources	0.959184	
32	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageStreamResolverUtils->MessageTypeResolver	0.959184	
33	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	DefaultParameterResolverFactory\$AnnotatedMetadataParameterResolver->Metadata	0.959184	
34	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageStreamResolverUtils->FluxUtils	0.959184	
35	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	SourceIdParameterResolverFactory <i>SourceIdParameterResolver</i> ->ContextResourceKey	0.959184	
36	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageStreamResolverUtils->MonoUtils	0.959184	
37	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MessageStreamResolverUtils->MessageStream	0.959184	
38	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	DefaultParameterResolverFactory\$AnnotatedMetadataParameterResolver->Message	0.959184	
39	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	MethodInvokingMessageHandlingMember->MessageStream\$Entry	0.959184	

Table 2c - Cyclic Dependencies Break Down - Backward Dependencies Only

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first. This table only contains the backward dependencies from the second participant of the cycle back to the first one that are the most promising to resolve.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic_Dependencies_Breakdown_BackwardOnly

	artifactName	packageName	dependentArtifactName	dependentPackageName	dependency forName
0	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core	HandlerAttributes<-SimpleHandlerAttributes
1	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	axon-messaging-5.0.3	org.axonframework.messaging.core	EventMessage<-SubscribableEventSource
2	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	EventHandler<-HandlerTypeResolver
3	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	CommandHandler<-HandlerTypeResolver
4	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	QueryHandler<-HandlerTypeResolver
5	axon-modelling-5.0.3	org.axonframework.modelling.annotation	axon-modelling-5.0.3	org.axonframework.modelling	TargetEntityIdMemberMismatchException<-PropertyBasedEntityIdResolver
6	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.transaction	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	TransactionManager<-TransactionalUnitOfWorkFactory
7	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	PooledStreamingEventProcessorsConfigurer<-EventProcessingConfigurer
8	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	PooledStreamingEventProcessorConfiguration<-EventProcessorModule
9	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.streaming.pooled	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	PooledStreamingEventProcessorModule<-EventProcessorModule
10	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	SubscribingEventProcessorConfiguration<-EventProcessorModule
11	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	SubscribingEventProcessorsConfigurer<-EventProcessingConfigurer
12	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.processing.subscribing	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	SubscribingEventProcessorModule<-EventProcessorModule
13	axon-common-5.0.3	org.axonframework.common.configuration	axon-common-5.0.3	org.axonframework.common.infra	Component\$Identifier<-JacksonComponentDescriptor
14	axon-common-5.0.3	org.axonframework.common.configuration	axon-common-5.0.3	org.axonframework.common.infra	Component<-JacksonComponentDescriptor
15	axon-common-5.0.3	org.axonframework.common.configuration	axon-common-5.0.3	org.axonframework.common.infra	Component<-FileSystemStyleComponentDescriptor
16	axon-common-5.0.3	org.axonframework.common.configuration	axon-common-5.0.3	org.axonframework.common.infra	Component\$Identifier<-FileSystemStyleComponentDescriptor
17	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	NoMoreInterceptors<-ChainedMessageHandlerInterceptorMember
18	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	NoMoreInterceptors<-AnnotatedHandlerInspector
19	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	MessageHandlerInterceptorMemberChain<-AnnotatedHandlerInspector
20	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	MessageInterceptingMember<-AnnotatedHandlerInspector
21	axon-messaging-5.0.3	org.axonframework.messaging.core.interception.annotation	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	MessageHandlerInterceptorMemberChain<-ChainedMessageHandlerInterceptorMember
22	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	EventProcessorControlService<-AxonServerConfigurationEnhancer
23	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	AxonServerEventStorageEngineFactory<-AxonServerConfigurationEnhancer
24	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.command	MetadataConverter<-CommandConverter
25	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.command	AxonServerConfiguration<-AxonServerCommandBusConnector
26	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.configuration	MessagingConfigurer<-EventProcessingConfigurer
27	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	axon-messaging-5.0.3	org.axonframework.messaging.core.sequencing	EventMessage<-ExtractionSequencingPolicy
28	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	axon-messaging-5.0.3	org.axonframework.messaging.core.sequencing	EventMessage<-SequentialPerAggregatePolicy
29	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	AxonServerRemoteQueryHandlingException<-ErrorCode
30	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	AxonServerQueryDispatchException<-ErrorCode
31	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	AxonServerNonTransientRemoteQueryHandlingException<-ErrorCode
32	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	AxonServerQueryBusConnector<-AxonServerConfigurationEnhancer
33	axon-modelling-5.0.3	org.axonframework.modelling.entity.annotation	axon-modelling-5.0.3	org.axonframework.modelling.annotation	AnnotatedEntityMetamodel<-EntityIdResolverDefinition
34	axon-modelling-5.0.3	org.axonframework.modelling.entity.annotation	axon-modelling-5.0.3	org.axonframework.modelling.annotation	AnnotatedEntityMetamodel<-AnnotationBasedEntityIdResolverDefinition
35	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	EntityChildMetamodel<-PolymorphicEntityMetamodelBuilder
36	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	EntityChildMetamodel<-EntityMetamodelBuilder
37	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	EntityChildMetamodel<-PolymorphicEntityMetamodelBuilder
38	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	EntityChildMetamodel<-ConcreteEntityMetamodel
39	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	axon-modelling-5.0.3	org.axonframework.modelling.entity	ChildAmbiguityException<-ConcreteEntityMetamodel

Interface Segregation Candidates

Well known from [Design Principles and Design Patterns by Robert C. Martin](#), the *Interface Segregation Principle* suggests that software components should have narrow, focused interfaces rather than large, general-purpose ones. The goal is to minimize the dependencies between components and increase modularity, flexibility, and maintainability.

Smaller, focused and purpose-driven interfaces

- make it easier to modify individual components without affecting the rest of the system.
- make it clearer which client is affected by which change.
- don't force their clients to depend on methods they don't need.
- reduce the scope of changes since a change to one component doesn't affect others.
- lead to a more loosely coupled architecture that is easier to understand and maintain.

Reference: [Analyze java package metrics in a graph database](#)

How to apply the results

If just one method of a type is used, especially in many places, then the result of this method can be used to call e.g. a method or construct an object instead of using the whole object and then just calling that single method.

If there are a couple of methods that are used for a distinct purpose, those could be factored out into a separate interface. The original type can extended/implement the new interface so that there are no breaking changes. Then all the callers, that use only this group of methods, can be changed to the new interface.

Table 4 - Top 40 most used combinations of methods

The following table shows the top 40 most used combinations of methods of larger types that might benefit from applying the *Interface Segregation Principle*. The whole table can be found in the CSV report `Candidates_for_Interface_Segregation`.

	fullDependentTypeName	declaredMethods	calledMethodNames	calledMethods	callerTypes
0	org.axonframework.messaging.core.unitofwork.ProcessingContext	32	[computeResourceIfAbsent]	1	7
1	org.axonframework.messaging.commandhandling.CommandBus	6	[dispatch]	1	7
2	org.axonframework.common.configuration.ComponentDefinition\$ComponentCreator	17	[createComponent]	1	5
3	org.axonframework.messaging.core.MessageStream\$Entry	7	[message]	1	5
4	org.axonframework.messaging.core.unitofwork.ProcessingContext	32	[withResource]	1	4
5	org.axonframework.messaging.eventhandling.EventMessage	19	[identifier, timestamp]	2	4
6	org.axonframework.messaging.eventhandling.EventMessage	19	[timestamp]	1	4
7	org.axonframework.messaging.core.MessageStream\$Empty	44	[cast]	1	3
8	org.axonframework.messaging.core.DelayedMessageStream	42	[create]	1	3
9	org.axonframework.messaging.eventhandling.EventMessage	19	[andMetadata]	1	3
10	org.axonframework.messaging.core.annotation.WrappedMessageHandlingMember	13	[handleSync]	1	3
11	org.axonframework.common.configuration.AbstractComponent	24	[describeTo]	1	2
12	org.axonframework.messaging.core.unitofwork.UnitOfWork	24	[execute]	1	2
13	org.axonframework.test.fixture.AxonTestThenMessage	22	[exception, exceptionSatisfies]	3	2
14	org.axonframework.messaging.commandhandling.CommandMessage	20	[priority, routingKey]	2	2
15	org.axonframework.messaging.commandhandling.CommandMessage	19	[withConvertedPayload]	1	2
16	org.axonframework.messaging.commandhandling.CommandMessage	19	[andMetadata]	1	2
17	org.axonframework.messaging.eventhandling.EventMessage	19	[withConvertedPayload]	1	2
18	org.axonframework.messaging.eventhandling.EventMessage	19	[identifier]	1	2
19	org.axonframework.messaging.eventstreaming.EventCriterion	12	[tags]	1	2
20	org.axonframework.messaging.eventstreaming.OrEventCriteria	12	[or]	1	2
21	org.axonframework.modelling.entity.ConcreteEntityMetamodel	11	[forEntityClass]	1	2
22	org.axonframework.modelling.entity.PolymorphicEntityMetamodel	11	[forSuperType]	1	2
23	org.axonframework.eventsourcing.eventstore.EventStore	9	[transaction]	1	2
24	org.axonframework.messaging.eventhandling.processing.streaming.token.store.ConfigToken	9	[get]	1	2
25	org.axonframework.messaging.core.MessageStream\$Entry	7	[map]	1	2
26	org.axonframework.conversion.ChainingContentTypeConverter	5	[registerConverter, canConvert]	2	2
27	org.axonframework.messaging.core.unitofwork.UnitOfWork\$UnitOfWorkProcessingContext	45	[on, commit, whenComplete, isCommitted, isCompleted, isStarted, isError, onError]	8	1
28	org.axonframework.messaging.core.EmptyMessageStream	44	[instance]	1	1
29	org.axonframework.eventsourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream	43	[hasNextAvailable, isCompleted]	2	1
30	org.axonframework.eventsourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream	42	[callback]	1	1
31	org.axonframework.messaging.core.DelayedMessageStream	42	[createSingle]	1	1
32	org.axonframework.messaging.core.MessageStream\$Single	42	[asCompletableFuture]	1	1
33	org.axonframework.messaging.core.MessageStream\$Single	42	[cast]	1	1
34	org.axonframework.messaging.core.MessageStream\$Single	42	[first]	1	1
35	org.axonframework.messaging.core.unitofwork.ProcessingContext	37	[updateResource, removeResource, putResourceIfAbsent, putResource, computeResourceIfAbsent]	6	1
36	org.axonframework.messaging.core.unitofwork.ProcessingContext	36	[updateResource, computeResourceIfAbsent]	2	1
37	org.axonframework.messaging.core.unitofwork.ProcessingContext	35	[computeResourceIfAbsent]	1	1
38	org.axonframework.messaging.core.unitofwork.ProcessingContext	34	[computeResourceIfAbsent]	1	1
39	org.axonframework.messaging.core.unitofwork.ProcessingContext	33	[putResourceIfAbsent, computeResourceIfAbsent]	2	1

Package Usage

Table 5 - Types that are used by multiple packages

This table shows the top 40 packages that are used by the highest number of different packages. The whole table can be found in the CSV report `List_types_that_are_used_by_many_different_packages` .

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
0	org.axonframework.messaging.core.unitofwork.ProcessingContext	ProcessingContext	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...	59
1	org.axonframework.common.annotation.Internal	Internal	[Type, File, Java, ByteCode, Annotation, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityCloseness, Mark4TopCentralityHarmonic, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent...	55
2	org.axonframework.messaging.core.Message	Message	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...	47
3	org.axonframework.common.infra.ComponentDescriptor	ComponentDescriptor	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityCloseness, Mark4TopCentralityHarmonic, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...	46
4	org.axonframework.messaging.core.MessageStream	MessageStream	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityCloseness, Mark4TopCentralityHarmonic, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWea...	36
5	org.axonframework.messaging.eventhandling.EventMessage	EventMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponen...	35
6	org.axonframework.messaging.core.MessageType	MessageType	[Type, File, Java, ByteCode, Record, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...	31
7	org.axonframework.messaging.core.QualifiedName	QualifiedName	[Type, File, Java, ByteCode, Record, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4T...	30
8	org.axonframework.common.configuration.Configuration	Configuration	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation14, Mark4TypeLouvainCommunity7...	29
9	org.axonframework.common.FutureUtils	FutureUtils	[Type, File, Java, Class, ByteCode, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity12, Mark...	28
10	org.axonframework.messaging.core.Context\$ResourceKey	Context\$ResourceKey	[Type, File, Java, Class, ByteCode, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityCloseness, Mark4TopCentralityHarmonic, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyCon...	25
11	org.axonframework.messaging.core.MessageStream\$Empty	MessageStream\$Empty	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity3, Mark4TypeKC...	25
12	org.axonframework.common.infra.DescribableComponent	DescribableComponent	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityCloseness, Mark4TopCentralityHarmonic, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...	24
13	org.axonframework.common.Assert	Assert	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityCloseness, Mark4TopCentralityHarmonic, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeKCoreDecompositi...	22
14	org.axonframework.common.configuration.ComponentRegistry	ComponentRegistry	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityCloseness, Mark4TopCentralityHarmonic, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation13, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut2, Mark4TypeHDBSCAN-1]	22
15	org.axonframework.common.BuilderUtils	BuilderUtils	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TypeLabelPropagation5, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity2, Mark4TypeKCoreDecomposition9, Mark4TypeMaximumKCut83, Mark4TypeHDBSCAN51, Mark4TopAnomalyHub]	21
16	org.axonframework.messaging.commandhandling.CommandMessage	CommandMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity9, Mark4TypeLeidenCommunity8, ...	20
17	org.axonframework.messaging.core.annotation.ParameterResolverFactory	ParameterResolverFactory	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity13, Mark4TypeLeidenCommunity1, Mark4TypeKCoreDecomposition9, Mark4TypeMaximumKCut2, Mark4TypeHDBSCAN-1]	20
18	org.axonframework.messaging.core.MessageStream\$Entry	MessageStream\$Entry	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0,	19

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
			Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity3, Mark4Ty...	
19	org.axonframework.common.configuration.ConfigurationEnhancer	ConfigurationEnhancer	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation17, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut48, Mark4TypeHDBSCAN-1]	18
20	org.axonframework.conversion.Converter	Converter	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity4, Mark4TypeCoreDecompo...	18
21	org.axonframework.messaging.core.MessageStream\$Single	MessageStream\$Single	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark...	18
22	org.axonframework.messaging.core.MessageTypeResolver	MessageTypeResolver	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity4, Mark4TypeCoreDecomposition10, ...	18
23	org.axonframework.messaging.core.conversion.MessageConverter	MessageConverter	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut24, Mark4TypeLocalClusteringCoefficient0.11333333333333333, Mark4TypeHDBSCAN-1]	17
24	org.axonframework.messaging.core.Metadata	Metadata	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity4, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut30, Ma...	16
25	org.axonframework.common.ObjectUtils	ObjectUtils	[Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity4, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut38, Mark4TypeHDBSCAN63]	16
26	org.axonframework.messaging.eventhandling.processing.streaming.token.TrackingToken	TrackingToken	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation39, Mark4TypeLouvainCommunity1...	15
27	org.axonframework.common.AxonConfigurationException	AxonConfigurationException	[Type, File, Java, Class, ByteCode, Throwable, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation27, Mark4TypeLouvainCommunity8, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut90, Mark4TypeHDBSCAN-1]	14
28	org.axonframework.messaging.commandhandling.CommandBus	CommandBus	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity9, Mark4TypeLeidenCommunity8, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut62, Mark4TypeHDBSCAN-1]	13
29	org.axonframework.common.ReflectionUtils	ReflectionUtils	[Type, File, Java, Class, ByteCode, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation19, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut45, Mark4TypeHDBSCAN53, Mark4TopAnomalyBottleneck, Mark4Top...	13
30	org.axonframework.messaging.commandhandling.CommandResultMessage	CommandResultMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity4, Mark4TypeCoreDecomposition10, ...	12
31	org.axonframework.common.configuration.ComponentDefinition	ComponentDefinition	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation11, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut89, Mark4TypeHDBSCAN-1]	12
32	org.axonframework.common.configuration.ComponentDefinition\$IncompleteComponentDefinition	ComponentDefinition\$IncompleteComponentDefinition	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation11, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut63, Mark4TypeHDBSCAN17]	12
33	org.axonframework.messaging.core.Context	Context	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCo...	12
34	org.axonframework.messaging.queryhandling.QueryMessage	QueryMessage	[Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation9, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity3, Mark4TypeCoreDecomposition10, ...	12
35	org.axonframework.common.AxonNonTransientException	AxonNonTransientException	[Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation2, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity5, Mark4TypeCoreDecomposition4, Mark4TypeMaximumKCut1, Mark4TypeLocalClusteringCoeff...	11
36	org.axonframework.common.configuration.ComponentBuilder	ComponentBuilder	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation13, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity6,	11

	fullQualifiedDependentTypeName	dependentTypeName	dependentTypeLabels	numberOfUsingPackages
			Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN81]	
37	org.axonframework.messaging.eventhandling.conversion.EventConverter	EventConverter	[Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity13, Mark4TypeLeidenCommunity8, Mark4TypeKCoreDecomposition9, Mark4TypeMaximumKCut75, Mark4TypeHDBSCAN-1]	11
38	org.axonframework.messaging.core.MessageHandlerInterceptor	MessageHandlerInterceptor	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation6, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity6, Mark4TypeKC...	11
39	org.axonframework.messaging.core.annotation.ParameterResolver	ParameterResolver	[Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity8, Mark4TypeLeidenCommunity1, Mark4TypeKCoreDecomposition7, Mark4TypeMaximumKCut1, Mark4TypeHDBSCAN27]	11

Table 6 - Packages that are used by multiple artifacts

This table shows the top 30 artifacts that only use a few (compared to all existing) packages of another artifact. The whole table can be found in the CSV report `ArtifactPackageUsage`.

	artifactName	dependentArtifactName	dependentPackages	dependentArtifactPackages	packageUsagePercentage	dependentFullQualifiedPackageNames	dependentPackageNames
0	axon-tracing-opentelemetry-5.0.3	axon-messaging-5.0.3	2	59	0.033898	[org.axonframework.messaging.core, org.axonframework.messaging.tracing]	[core, tracing]
1	axon-tracing-opentelemetry-5.0.3	axon-common-5.0.3	1	15	0.066667	[org.axonframework.common]	[common]
2	axon-metrics-micrometer-5.0.3	axon-messaging-5.0.3	7	59	0.118644	[org.axonframework.messaging.commandhandling, org.axonframework.messaging.core, org.axonframework.messaging.eventhandling, org.axonframework.messaging.queryhandling, org.axonframework.messaging.monitoring, org.axonframework.messaging.eventhandling.processing, org.axonframework.messaging.monitori...	[commandhandling, core, eventhandling, queryhandling, monitoring, processing, configuration]
3	axon-metrics-micrometer-5.0.3	axon-common-5.0.3	2	15	0.133333	[org.axonframework.common.configuration, org.axonframework.common]	[configuration, common]
4	axon-test-5.0.3	axon-eventsourcing-5.0.3	1	7	0.142857	[org.axonframework.eventsourcing.eventstore]	[eventstore]
5	axon-server-connector-5.0.3	axon-modelling-5.0.3	1	7	0.142857	[org.axonframework.modelling]	[modelling]
6	axon-server-connector-5.0.3	axon-eventsourcing-5.0.3	1	7	0.142857	[org.axonframework.eventsourcing.eventstore]	[eventstore]
7	axon-metrics-micrometer-5.0.3	axon-spring-boot-autoconfigure-5.0.3	1	7	0.142857	[org.axonframework.extension.springboot.autoconfig]	[autoconfig]
8	axon-test-5.0.3	axon-messaging-5.0.3	9	59	0.152542	[org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.core, org.axonframework.messaging.monitoring, org.axonframework.messaging.commandhandling, org.axonframework.messaging.core.annotation, org.axonframework.messaging.eventhandling, org.axonframework.messaging.eventhandling.p...	[unitofwork, core, monitoring, commandhandling, annotation, eventhandling, token, conversion, eventstreaming]
9	axon-spring-boot-autoconfigure-5.0.3	axon-test-5.0.3	1	6	0.166667	[org.axonframework.test.server]	[server]
10	eventsourcing-5.0.3	axon-conversion-5.0.3	1	5	0.200000	[org.axonframework.conversion]	[conversion]
11	axon-conversion-5.0.3	axon-conversion-5.0.3	1	5	0.200000	[org.axonframework.conversion]	[conversion]
12	axon-modelling-5.0.3	axon-conversion-5.0.3	1	5	0.200000	[org.axonframework.conversion]	[conversion]
13	axon-conversion-5.0.3	axon-common-5.0.3	3	15	0.200000	[org.axonframework.common, org.axonframework.common.infra, org.axonframework.common.annotation]	[common, infra, annotation]
14	axon-spring-boot-autoconfigure-5.0.3	axon-server-connector-5.0.3	1	5	0.200000	[org.axonframework.axonserver.connector]	[connector]
15	axon-update-5.0.3	axon-common-5.0.3	3	15	0.200000	[org.axonframework.common.annotation, org.axonframework.common, org.axonframework.common.configuration]	[annotation, common, configuration]
16	axon-modelling-5.0.3	axon-messaging-5.0.3	13	59	0.220339	[org.axonframework.messaging.queryhandling.configuration, org.axonframework.messaging.core.configuration, org.axonframework.messaging.commandhandling.configuration, org.axonframework.messaging.eventhandling, org.axonframework.messaging.core, org.axonf...	[configuration, commandhandling, eventhandling, core, unitofwork, annotation, conversion, reflection]
17	axon-test-5.0.3	axon-common-5.0.3	4	15	0.266667	[org.axonframework.common, org.axonframework.common.infra, org.axonframework.common.annotation, org.axonframework.common.configuration]	[common, infra, annotation, configuration]
18	axon-server-connector-5.0.3	axon-messaging-5.0.3	16	59	0.271186	[org.axonframework.messaging.core.unitofwork, org.axonframework.messaging.eventhandling.processing.streaming.token, org.axonframework.messaging.eventhandling.processing, org.axonframework.messaging.eventhandling.conversion, org.axonframework.messaging.eventstreaming, org.axonframework.messaging...	[unitofwork, token, processing, conversion, eventstreaming, streaming, subscribing, store, segmenting, core, eventhandling, distributed, queryhandling, commandhandling]
19	axon-eventsourcing-5.0.3	axon-messaging-5.0.3	16	59	0.271186	[org.axonframework.messaging.core.configuration, org.axonframework.messaging.eventhandling, org.axonframework.messaging.core.interception, org.axonframework.messaging.commandhandling, org.axonframework.messaging.core.annotation, org.axonframework.messaging.eventhandling.conversion, org.axonframe...	[configuration, eventhandling, interception, commandhandling, annotation, conversion, core, token, eventstreaming, unitofwork, transaction]
20	axon-spring-boot-autoconfigure-5.0.3	axon-messaging-5.0.3	16	59	0.271186	[org.axonframework.messaging.core.annotation, org.axonframework.messaging.commandhandling.distributed, org.axonframework.messaging.core.interception, org.axonframework.messaging.eventhandling.processing.streaming.token.store.jpa, org.axonframework.messaging.queryhandling.distributed, org.axonfra...	[annotation, distributed, interception, jpa, conversion, timeout, store, core, eventhandling, transaction, correlation, queryhandling, commandhandling]
21	axon-spring-boot-autoconfigure-5.0.3	axon-eventsourcing-5.0.3	2	7	0.285714	[org.axonframework.eventsourcing.eventstore.jpa, org.axonframework.eventsourcing.eventstore]	[jpa, eventstore]
22	axon-spring-boot-autoconfigure-5.0.3	axon-common-5.0.3	5	15	0.333333	[org.axonframework.common, org.axonframework.common.jdbc, org.axonframework.common.jpa, org.axonframework.common.configuration, org.axonframework.common.annotation]	[common, jdbc, jpa, configuration, annotation]
23	axon-modelling-5.0.3	axon-common-5.0.3	5	15	0.333333	[org.axonframework.common, org.axonframework.common.configuration, org.axonframework.common.infra, org.axonframework.common.annotation, org.axonframework.common.property]	[common, configuration, infra, annotation, property]
24	axon-server-connector-5.0.3	axon-common-5.0.3	6	15	0.400000	[org.axonframework.common.configuration, org.axonframework.common.infra, org.axonframework.common.annotation, org.axonframework.common.lifecycle, org.axonframework.common.util]	[configuration, infra, annotation, common, lifecycle, util]
25	axon-messaging-5.0.3	axon-conversion-5.0.3	2	5	0.400000	[org.axonframework.conversion, org.axonframework.conversion.jackson]	[conversion, jackson]
26	axon-eventsourcing-5.0.3	axon-common-5.0.3	8	15	0.533333	[org.axonframework.common, org.axonframework.common.annotation, org.axonframework.common.configuration, org.axonframework.common.infra, org.axonframework.common.tx, org.axonframework.common.jpa, org.axonframework.common.jdbc, org.axonframework.common.io]	[common, annotation, configuration, infra, tx, jpa, jdbc, io]
27	axon-common-5.0.3	axon-common-5.0.3	8	15	0.533333	[org.axonframework.common.infra, org.axonframework.common.lifecycle, org.axonframework.common.annotation, org.axonframework.common, org.axonframework.common.configuration, org.axonframework.common.tx, org.axonframework.common.function, org.axonframework.common.io]	[infra, lifecycle, annotation, common, configuration, tx, function, io]
28	axon-spring-boot-autoconfigure-5.0.3	axon-update-5.0.3	3	5	0.600000	[org.axonframework.update.detection, org.axonframework.update.configuration, org.axonframework.update]	[detection, configuration, update]
29	axon-metrics-micrometer-5.0.3	axon-metrics-micrometer-5.0.3	2	3	0.666667	[org.axonframework.extension.metrics.micrometer, org.axonframework.extension.metrics.micrometer.reservoir]	[micrometer, reservoir]

Table 7 - Types that are used by multiple artifacts

This table shows the top 30 types that only use a few (compared to all existing) types of another artifact. The whole table can be found in the CSV report `ClassesPerPackageUsageAcrossArtifacts`.

	artifactName	dependentArtifactName	packageName	dependentPackage.fqn	dependentTypes	dependentPackageTypes	typeUsagePercentage	
0	axon-eventsourcing-5.0.3	axon-messaging-5.0.3	org.axonframework.eventsourcing.configuration	org.axonframework.messaging.core	1	80	0.012500	
1	axon-test-5.0.3	axon-messaging-5.0.3	org.axonframework.test.matchers	org.axonframework.messaging.core	1	80	0.012500	
2	axon-modelling-5.0.3	axon-messaging-5.0.3	org.axonframework.modelling.repository	org.axonframework.messaging.core	1	80	0.012500	
3	axon-eventsourcing-5.0.3	axon-messaging-5.0.3	org.axonframework.eventsourcing.configuration	org.axonframework.messaging.core.annotation	1	51	0.019608	[org.axonfr
4	axon-eventsourcing-5.0.3	axon-common-5.0.3	org.axonframework.eventsourcing.annotation	org.axonframework.common.configuration	1	46	0.021739	
5	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core	org.axonframework.common.configuration	1	46	0.021739	
6	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core.unitofwork	org.axonframework.common.configuration	1	46	0.021739	[org.axonfr
7	axon-modelling-5.0.3	axon-common-5.0.3	org.axonframework.modelling.entity.annotation	org.axonframework.common.configuration	1	46	0.021739	
8	axon-eventsourcing-5.0.3	axon-common-5.0.3	org.axonframework.eventsourcing.annotation.reflection	org.axonframework.common.configuration	1	46	0.021739	
9	axon-metrics-micrometer-5.0.3	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.metrics.micrometer.springboot	org.axonframework.extension.springboot.autoconfig	1	41	0.024390	[org.axonfr
10	axon-tracing-opentelemetry-5.0.3	axon-messaging-5.0.3	org.axonframework.extension.tracing.opentelemetry	org.axonframework.messaging.core	2	80	0.025000	
11	axon-spring-boot-autoconfigure-5.0.3	axon-common-5.0.3	org.axonframework.extension.springboot.autoconfig	org.axonframework.common	1	34	0.029412	
12	axon-update-5.0.3	axon-common-5.0.3	org.axonframework.update	org.axonframework.common	1	34	0.029412	
13	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.monitoring	org.axonframework.common	1	34	0.029412	
14	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core.interception.annotation	org.axonframework.common	1	34	0.029412	
15	axon-conversion-5.0.3	axon-common-5.0.3	org.axonframework.conversion.avro	org.axonframework.common	1	34	0.029412	
16	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core.unitofwork.annotation	org.axonframework.common	1	34	0.029412	
17	axon-test-5.0.3	axon-common-5.0.3	org.axonframework.test.util	org.axonframework.common	1	34	0.029412	
18	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.queryhandling.tracing	org.axonframework.common	1	34	0.029412	
19	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core.configuration.reflection	org.axonframework.common	1	34	0.029412	
20	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.eventstreaming	org.axonframework.common	1	34	0.029412	
21	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.eventhandling.processing	org.axonframework.common	1	34	0.029412	
22	axon-test-5.0.3	axon-common-5.0.3	org.axonframework.test.extension	org.axonframework.common	1	34	0.029412	
23	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.eventhandling.tracing	org.axonframework.common	1	34	0.029412	
24	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.commandhandling.tracing	org.axonframework.common	1	34	0.029412	
25	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core.sequencing	org.axonframework.common	1	34	0.029412	
26	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.core.timeout	org.axonframework.common	1	34	0.029412	
27	axon-test-5.0.3	axon-common-5.0.3	org.axonframework.test.server	org.axonframework.common	1	34	0.029412	
28	axon-messaging-5.0.3	axon-common-5.0.3	org.axonframework.messaging.monitoring.configuration	org.axonframework.common	1	34	0.029412	
29	axon-tracing-opentelemetry-5.0.3	axon-common-5.0.3	org.axonframework.extension.tracing.opentelemetry	org.axonframework.common	1	34	0.029412	

Table 8 - Duplicate package names across artifacts

This table shows the top 30 duplicate package names across artifacts. They are ordered by the number of duplicates descending.

This might lead to confusion, makes importing more error prone and might even lead to duplicate classes where only one of them will be loaded by the class loader. If a package is named the same way in two or more artifacts this even allows another artifact to access package protected classes, methods or members which might not be intended.

The whole table can be found in the CSV report `DuplicatePackageNamesAcrossArtifacts`.

packageName	duplicates	artifactNames
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Table 9 - Annotated elements

This table shows 30 most used Java Annotations including some examples where they are used.

dependency.fileDistanceAsFewestChangeDirectoryCommands	numberOfDependencies	numberOfDependencyUsers	numberOfDependencyProviders	examples	
0	0	2125	881	921	[/axon-eventsourcing-5.0.3.jar uses /axon-conversion-5.0.3.jar, /axon-spring-boot-autoconfigure-5.0.3.jar uses /axon-conversion-5.0.3.jar, /axon-messaging-5.0.3.jar uses /axon-conversion-5.0.3.jar, /axon-modelling-5.0.3.jar uses /axon-conversion-5.0.3.jar]
1	1	100	86	41	[/org/axonframework/conversion/jackson2 uses /org/axonframework/conversion, /org/axonframework/conversion/avro uses /org/axonframework/conversion, /org/axonframework/conversion/jackson uses /org/axonframework/conversion, /org/axonframework/conversion/converter uses /org/axonframework/conversion]
2	2	2138	631	420	[/org/axonframework/conversion/avro/AvroConverter.class uses /org/axonframework/conversion/ChainingContentTypeConverter.class, /org/axonframework/conversion/jackson/JacksonConverter.class uses /org/axonframework/conversion/ChainingContentTypeConverter.class, /org/axonframework/conversion/jackson...
3	4	2104	679	322	[/org/axonframework/messaging/eventhandling/conversion uses /org/axonframework/conversion, /org/axonframework/messaging/core/configuration uses /org/axonframework/conversion, /org/axonframework/messaging/commandhandling/gateway uses /org/axonframework/conversion, /org/axonframework/messaging/eve...